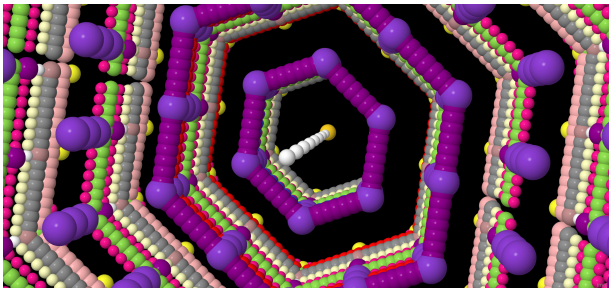
# Representation and Construction of Perfects

*An Illustrative Exposition of Fermatś Last Theorem and Bealś Conjecture*



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**Abstract**

Having two or more identical sources with which to form the perfect constrains construction to two or more layers at a time which results in the exclusion of the apex.

# 1 Problems

$$S \begin{cases} A^2 + B^2 = C^2 & \text{Pythagorean Theorem} \\ A^N + B^N = C^N & \text{Fermat's Last Theorem} \\ A^X + B^Y = C^Z & \text{Beal's Conjecture} \end{cases}$$

# 2 Computational Algorithm

$$H_\Delta \begin{cases} 0 & \theta \text{ zero} \\ \Delta(\theta, \beta, \delta) & \theta, \beta \text{ and } \delta \text{ whole} \end{cases}$$

$$\Delta(\theta, \beta, \delta) = (\beta + \delta)\Delta((\theta - 1), \beta, \delta) + \delta \sum_{r=1}^{\beta-1} \Delta((\theta - 1), r, \delta)$$

$$\begin{aligned} C_\theta &= P^\theta - P^{(\theta-2)} \\ C_\theta &= P^{(\theta-2)}(P^2 - 1) \\ C_\theta &= P^{(\theta-2)}(P - 1)(P + 1) \\ C_\theta &= P^{(\theta-3)}(P - 1)(P)(P + 1) \end{aligned}$$

$$C_\theta \text{ mod } 6 \equiv 0$$

$$C_\theta \quad \text{has six identical parts when } \theta > 2$$

$$\begin{aligned} L0_\beta &= \Delta(\theta, \beta + 0, \delta = 1) - \Delta(\theta - 2, \beta + 0, \delta = 1) \\ L1_\beta &= \Delta(\theta, \beta + 1, \delta = 1) - \Delta(\theta - 2, \beta + 1, \delta = 1) \\ L2_\beta &= \Delta(\theta, \beta + 2, \delta = 1) - \Delta(\theta - 2, \beta + 2, \delta = 1) \end{aligned}$$

R is the differences in adjacent layers which characterize the ring

$$\begin{aligned} R1_\beta &= (L1_\beta - L0_\beta) \\ R2_\beta &= (L2_\beta - L1_\beta) \end{aligned}$$

$$\epsilon_\psi = (R2_\beta - R1_\beta)$$

$$\epsilon_\theta = \begin{cases} 1 & \theta = 2(flat) \\ \epsilon_\psi/6 & \theta > 2 \end{cases}$$

Equations shown here define the $H_p^y e$ mechanism. See Tables 3 6 7.

The mechanism generates the complete set of differences for perfects of all orders. The difference of two perfects is an additive method. The recursive function uses only addition. Perfects are formed from the subpolynomial clusters, $C_\theta$. The cluster is shown to have a 6 symmetry(6s). The epsilon weights, $\epsilon_\psi$ are 6s characteristics of a layer which define the weight of the rings in the cluster. $\epsilon_\psi$ is the weight of single strata in a ring of the layer.

All closed form polynominal equations for $\epsilon_\psi$ are derived from the table. Tables 1 2 describes the growth of $\epsilon_\psi$. The table has two sides. The $(\beta - 1)$ side has an alternating series of terms. The $(\beta + 1)$ side has an increasing series of terms. The rows in table are the coefficients of the polynomial sum. Alternating columns cancel out.

| | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 1 | 0 | | | | | | | | | | | |
| 1 | 1 | 1 | | | | | | | | | | | |
| 2 | 1 | 2 | 0 | | | | | | | | | | |
| 3 | 1 | 3 | 2 | | | | | | | | | | |
| 4 | 1 | 4 | 5 | 2 | | | | | | | | | |
| 5 | 1 | 5 | 9 | 7 | 2 | | | | | | | | |
| 6 | 1 | 6 | 14 | 16 | 9 | 2 | | | | | | | |
| 7 | 1 | 7 | 20 | 30 | 25 | 11 | 2 | | | | | | |
| 8 | 1 | 8 | 27 | 50 | 55 | 36 | 13 | 2 | | | | | |
| 9 | 1 | 9 | 35 | 77 | 105 | 91 | 49 | 15 | 2 | | | | |
| 10 | 1 | 10 | 44 | 112 | 182 | 196 | 140 | 64 | 17 | 2 | | | |
| 11 | 1 | 11 | 54 | 156 | 294 | 378 | 336 | 204 | 81 | 19 | 2 | | |
| 12 | 1 | 12 | 65 | 210 | 450 | 672 | 714 | 540 | 285 | 100 | 21 | 2 | |
| 13 | 1 | 13 | 77 | 275 | 660 | 1122 | 1386 | 1254 | 825 | 385 | 121 | 23 | 2 |

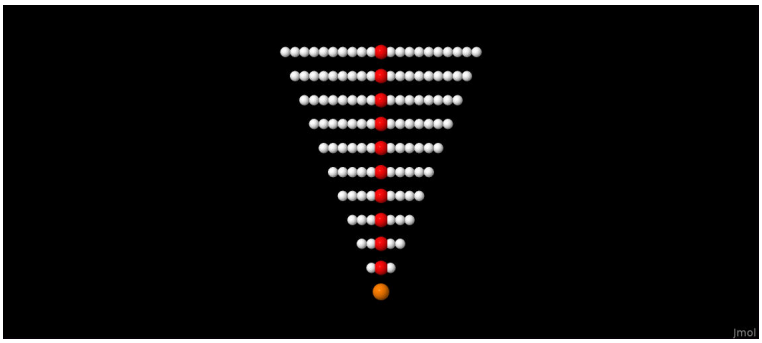
Table 1: $\Delta\ \epsilon$ growth L2

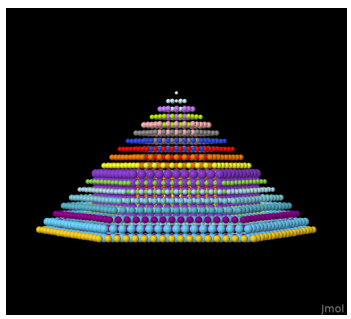
| | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 1 | 0 | | | | | | | | | | | |
| 1 | 1 | -1 | | | | | | | | | | | |
| 2 | 1 | -2 | 0 | | | | | | | | | | |
| 3 | 1 | -3 | 2 | | | | | | | | | | |
| 4 | 1 | -4 | 5 | -2 | | | | | | | | | |
| 5 | 1 | -5 | 9 | -7 | 2 | | | | | | | | |
| 6 | 1 | -6 | 14 | -16 | 9 | -2 | | | | | | | |
| 7 | 1 | -7 | 20 | -30 | 25 | -11 | 2 | | | | | | |
| 8 | 1 | -8 | 27 | -50 | 55 | -36 | 13 | -2 | | | | | |
| 9 | 1 | -9 | 35 | -77 | 105 | -91 | 49 | -15 | 2 | | | | |
| 10 | 1 | -10 | 44 | -112 | 182 | -196 | 140 | -64 | 17 | -2 | | | |
| 11 | 1 | -11 | 54 | -156 | 294 | -378 | 336 | -204 | 81 | -19 | 2 | | |
| 12 | 1 | -12 | 65 | -210 | 450 | -672 | 714 | -540 | 285 | -100 | 21 | -2 | |
| 13 | 1 | -13 | 77 | -275 | 660 | -1122 | 1386 | -1254 | 825 | -385 | 121 | -23 | 2 |

Table 2: $\Delta\ \epsilon$ growth L0

$$\epsilon_3 = 6\beta$$
$$\epsilon_4 = 12\beta^2$$
$$\epsilon_5 = 20\beta^3 + 4\beta$$
$$\epsilon_6 = 30\beta^4 + 18\beta^2$$
$$\epsilon_7 = 42\beta^5 + 50\beta^3 + 4\beta$$
$$\epsilon_8 = 56\beta^6 + 110\beta^4 + 26\beta^2$$

## 3 Representation and Construction



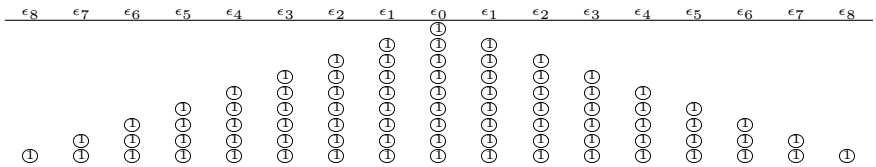

The perfect $\Delta^2$ can be represented as a triangular arrangement of points. The perfect $\Delta^3$ $\Delta^4 \ldots$ can be modeled in a hexagonal pyramid arrangement of points. The dimension or height of the perfect is the count of points in the center along with the apex; the $\epsilon_0$ column. The number of layers is the count of points in the $C_1$ center.

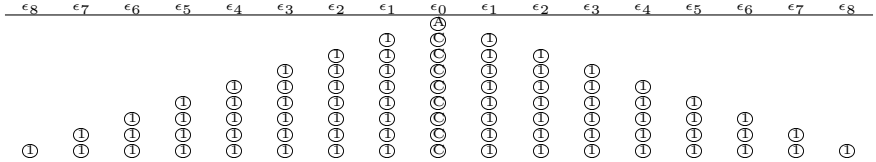
The 'volume' of a perfect is the summation of row elements, $\epsilon^\theta$, in the layer weight table. 3

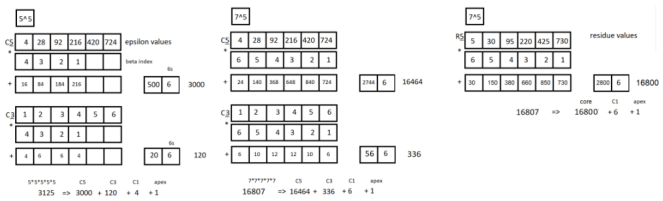
$$V(\Delta^\theta_\beta) = \sum_{x=1}^{\beta} LW(\theta, x)$$

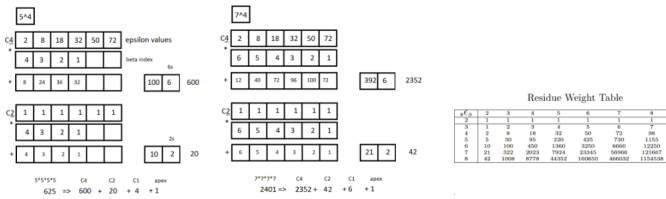
The 'volume' of the 3p wedge is the summation of row elements, $\epsilon^\theta$, in the residue weight table. 7

$$V(C^\theta_\beta) = \sum_{x=1}^{\beta-1}((\beta - x)RW(\theta, x))$$



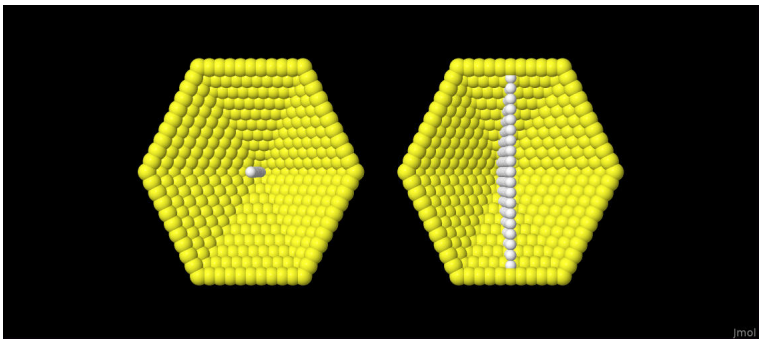



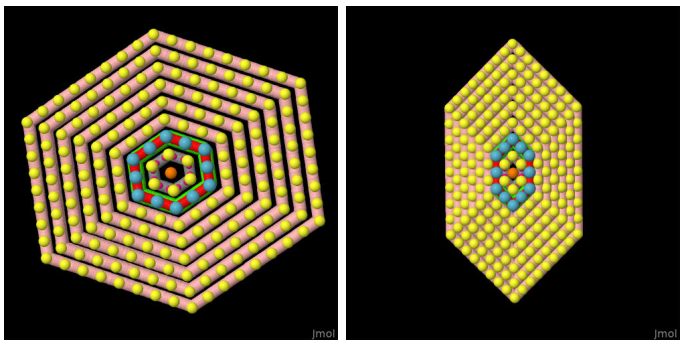




Residue Weight Table

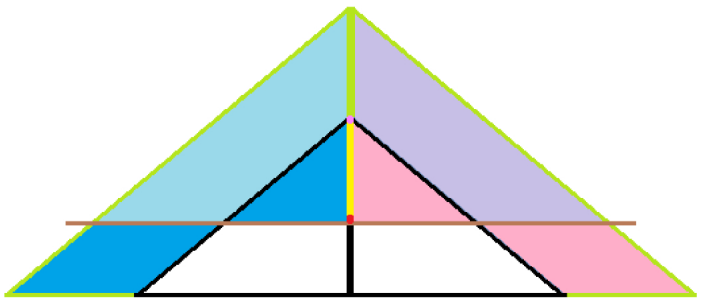
| $\epsilon^n$ $\beta^n$ | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|---|---|---|
| 2 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |
| 3 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
| 4 | 2 | 8 | 18 | 32 | 50 | 72 | 98 | 128 |
| 5 | 5 | 30 | 95 | 220 | 425 | 730 | 1155 | 1720 |
| 6 | 10 | 100 | 450 | 1360 | 3250 | 6660 | 12250 | 20800 |
| 7 | 21 | 322 | 2023 | 7924 | 23345 | 56966 | 121667 | 235368 |
| 8 | 42 | 1008 | 8778 | 44352 | 160650 | 466032 | 1154538 | 2542848 |

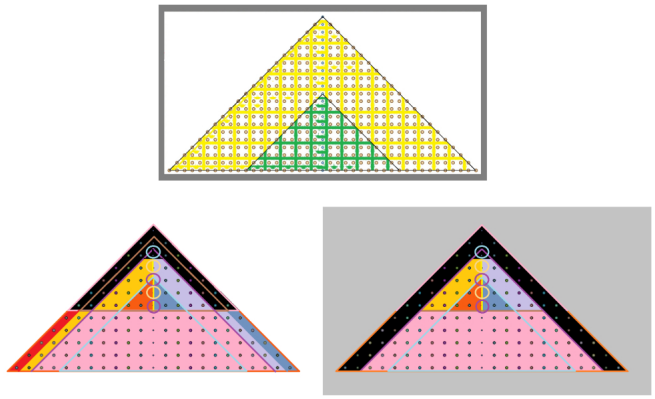




An odd perfect $\Delta^O$ has axial symmetry around the $C_1$ center. An even perfect $\Delta^E$ has a plane of symmetry about the $C_2$ center. The $C_2$ cluster is 2s around the $C_1$ center for bilateral symmetry. The odd orders are transformed into the same bilateral representation. Each 6s 3d wedge is represented as two 3p $\epsilon$ units around the $C_1$ center.

From the diagrams it can be seen that the difference operation is two sided and produces two symmetric parts which are avialable for rearrangement into the perfect $\Delta_A = \Delta_C$ - $\Delta_B$.





There are three obstacles one must hurdle in order to rearrange points after the difference operation. First the 6s3p left and right core pieces are formed from 3p components, then the center must be formed from 1p units and finally an apex placed.

One stipulates to the first two of these conditions. Constructing the center requires that $\epsilon^{\theta}$ units are decomposed into 1p units. The constraint of left and right sources force the center to be even. The center is formed from points in pairs. The newly forming object must be built two layers at a time. A 3p6s core can be formed for some measure of $\Delta^B$ and $\Delta^C$. An even center may be placed, however infrequent. But finally to place the apex is a challenge. The apex can not be placed atop the $\epsilon_0$ column.
A 2s3p core plus even constrained center plus apex results in an odd number of components.

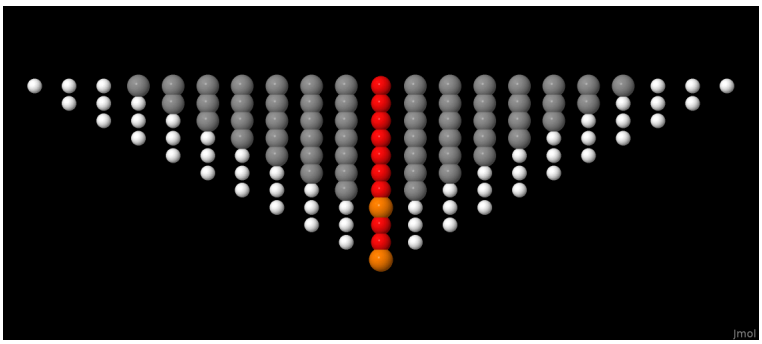
$$\Delta^{odd} = 1 + C_1 + C_3 + C_5 + \dots$$
$$\Delta^{even} = 1 + C_2 + C_4 + C_6 + \dots$$

$$\Delta^{odd} = apex + center + C_3 + C_5 + \dots$$
$$\Delta^{even} = apex + center + twotriangles + C_4 + C_6 + \dots$$

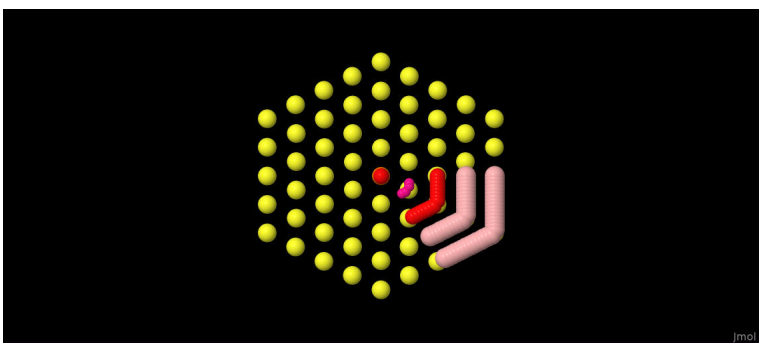
The difference operation[1] removes the apexs from both objects, ΔC and ΔB, prior to making the divet. A fulcrum is left in the divet. The missing divet is a constructable apex-less $\Delta_D$ similiar but smaller then ΔB. The divet is replenished with 3p points from the objects outer edges and/or c2 center difference

[1] in FLT case where $C^N - B^N = A^N$ and $A < B < C$

for even order. Some 3p points are fractured to fill the 1p center(two layers at a time). With divet and center complete the object forms an apex-less perfect. $\Delta_C - \Delta_B = \Delta_A - 1$. At this point Fermat's Last Theorem is shown by inspection.




The corner wedge arrangement is not bi-laterally symmetric but does maintain balance, there is a hex-cubic form, which when removed forms a divet in the object.




It is not possible to formulate a difference for all variables of $C^Z - B^Y = A^X$. It is possible to perfom an intersection of two objects. This leads to 'negative' space. To satisfy Beal's conjecture one must combine two objects.

Given the two objects, $\Delta_A$ and $\Delta_B$ form $\Delta_C$. Steps to construct perfect; Seperate the apex and c1 centers from both. Make two identical piles left and right from clusters of A and B. Make a minimal center.

$$C1_{init} = (a + b) \mod 6$$
$$R_{frac} = (a + b) - C1_{init}$$

Distribute $R_{frac}$ equally into the left 3p and right 3p piles.

When $C1_{init}$ is 1,2,3,4 or 5, one proceeds by forming the perfect $\Delta_C^X$ where X is the arbitrary order. When $C1_{init}$ is zero, one forms perfect after seeding with one of the apexes.

Seeded with 1, $\Delta_2^X$, $\Delta_3^X$, $\Delta_4^X$, $\Delta_5^X$ or $\Delta_6^X$ construction proceeds adding two layers at a time until piles exhausted. Every layer added to the growing form completes the object, a larger perfect. For any value of order X, the same holds true.

Construction of a perfect is not exact when two identical processes are used to build object. The two processes, one for each pile, perform after the initial configuration is seeded. There is always an even number of components to be utilized after seeding. The left and right piles, two sided. The extra 1p apex is not used. $\Delta_A + \Delta_B = \Delta_C + 1$. The chances of forming a 2s3p core is very small, filling the center even smaller. One succeeds in forming the perfect when piles are exhausted and the final two layers complete but fails to use all the points.

## 4 Summary

The question, 'How close does one get when reforming perfects?', has a minimum of one.

$\Delta_A + \Delta_B = \Delta_C + 1$. When adding perfects together, an extra apex.

$\Delta_C - \Delta_B = \Delta_A - 1$. When performing a difference, a missing apex.

| $\psi$ | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |
| 2 | 1 | 3 | 5 | 7 | 9 | 11 | 13 | 15 | 17 |
| 3 | 1 | 7 | 19 | 37 | 61 | 91 | 127 | 169 | 217 |
| 4 | 1 | 15 | 65 | 175 | 369 | 671 | 1105 | 1695 | 2465 |
| 5 | 1 | 31 | 211 | 781 | 2101 | 4651 | 9031 | 15961 | 26281 |
| 6 | 1 | 63 | 665 | 3367 | 11529 | 31031 | 70993 | 144495 | 269297 |
| 7 | 1 | 127 | 2059 | 14197 | 61741 | 201811 | 543607 | 1273609 | 2685817 |
| 8 | 1 | 255 | 6305 | 58975 | 325089 | 1288991 | 4085185 | 11012415 | 26269505 |

Table 3: Layer Weights the difference of two perfects $\Delta^{\psi}_{N+1} - \Delta^{\psi}_{N}$

| $\psi$ 2 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 |
| 2 | 4 | 8 | 12 | 16 | 20 | 24 | 28 | 32 | 36 |
| 3 | 8 | 26 | 56 | 98 | 152 | 218 | 296 | 386 | 488 |
| 4 | 16 | 80 | 240 | 544 | 1040 | 1776 | 2800 | 4160 | 5904 |
| 5 | 32 | 242 | 992 | 2882 | 6752 | 13682 | 24992 | 42242 | 672325 |
| 6 | 64 | 728 | 4032 | 14896 | 42560 | 102024 | 215488 | 413792 | 7378566 |
| 7 | 128 | 2186 | 16256 | 75938 | 263552 | 745418 | 1817216 | 3959426 | 79028487 |
| 8 | 256 | 6560 | 65280 | 384064 | 1614080 | 5374176 | 15097600 | 37281920 | 832227848 |

Table 4: Layer Weights the difference of two perfects $\Delta^{\psi}_{N+2} - \Delta^{\psi}_{N}$

| $\psi$ 3 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 |
| 2 | 9 | 15 | 21 | 27 | 33 | 39 | 45 | 51 | 57 |
| 3 | 27 | 63 | 117 | 189 | 279 | 387 | 513 | 657 | 819 |
| 4 | 81 | 255 | 609 | 1215 | 2145 | 3471 | 5265 | 7599 | 10545 |
| 5 | 243 | 1023 | 3093 | 7533 | 15783 | 29643 | 51273 | 83193 | 128283 |
| 6 | 729 | 4095 | 15561 | 45927 | 113553 | 246519 | 484785 | 882351 | 1509417 |
| 7 | 2187 | 16383 | 77997 | 277749 | 807159 | 2019027 | 4503033 | 9176457 | 17390019 |
| 8 | 6561 | 65535 | 390369 | 1673055 | 5699265 | 16386591 | 41367105 | 94235199 | 197581665 |

Table 5: Layer Weights the difference of two perfects $\Delta^{\psi}_{N+3} - \Delta^{\psi}_{N}$

| $\epsilon_\psi$ | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|---|---|---|
| 2 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |
| 3 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
| 4 | 2 | 8 | 18 | 32 | 50 | 72 | 98 | 128 |
| 5 | 4 | 28 | 92 | 216 | 420 | 724 | 1148 | 1712 |
| 6 | 8 | 92 | 432 | 1328 | 3200 | 6588 | 12152 | 20672 |
| 7 | 16 | 292 | 1928 | 7704 | 22920 | 56236 | 120512 | 233648 |
| 8 | 32 | 908 | 8328 | 42992 | 157400 | 459372 | 1142288 | 2522048 |

Table 6: $\epsilon$ weights single strata weight units for rings

| $\epsilon_\psi$ | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|---|---|---|
| 2 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |
| 3 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
| 4 | 2 | 8 | 18 | 32 | 50 | 72 | 98 | 128 |
| 5 | 5 | 30 | 95 | 220 | 425 | 730 | 1155 | 1720 |
| 6 | 10 | 100 | 450 | 1360 | 3250 | 6660 | 12250 | 20800 |
| 7 | 21 | 322 | 2023 | 7924 | 23345 | 56966 | 121667 | 235368 |
| 8 | 42 | 1008 | 8778 | 44352 | 160650 | 466032 | 1154538 | 2542848 |

Table 7: Residue Weight : sum of $\epsilon$ weights for all strata that form cluster